

ORCA: Automated Brain Tumor Segmentation Using Deep Learning

A 3D Attention U-Net Approach for Automated Glioma Segmentation in Multi-Modal MRI

Submitted by

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Project Orca: Automated Brain Tumor Segmentation Using Deep Learning

Abstract

Brain tumors, particularly high-grade gliomas, pose significant challenges in clinical care. Reliable segmentation of tumor subregions in multi-parametric magnetic resonance imaging (mpMRI) is pivotal for grading, surgical resection planning, radiotherapy targeting, and treatment response monitoring. Manual delineation remains time-intensive, inconsistent across observers, and operationally inefficient in busy clinical environments.

Project Orca introduces a fully automated, AI-driven volumetric tumor segmentation pipeline leveraging **3D Attention U-Net** architecture trained on the **BraTS 2023** dataset. This model extracts three clinically relevant sub-regions: **Whole Tumor (WT)**, **Tumor Core (TC)**, and **Enhancing Tumor (ET)**. The system integrates reproducible preprocessing, optimized training workflow, augmentation strategies, deployment readiness, comprehensive evaluation, and real-world workflow benefits. The report includes extensive analysis, system deployment architecture, benchmark comparisons, and clinical interpretation principles.

This report incorporates milestones covering dataset curation, exploratory data analysis (EDA), architectural advancement, training optimization, error categorization, user-centric application workflows, model packaging, and performance benchmarking. The final system aims to support radiologists and neuro-oncologists with faster decision-making, quantitative indicators, and reliable volumetric assessments.

1. Introduction

1.1 Clinical Motivation - Why Brain Tumor Segmentation Matters

Brain tumors disrupt neural functions, and treatment decisions are highly dependent on precise knowledge of tumor location and spatial invasiveness. MRI, as a modality,

enables superior visualization of soft tissue contrast. Yet, clinical segmentation faces major limitations:

- **Long segmentation time:** 30-90 minutes per case
- **Intra- and inter-observer variability**
- **Knowledge dependency on clinical experience**
- **Difficulties in monitoring small, progressive change**

Advanced segmentation methods allow:

- Improved surgical precision (clean margins, maximal safe resection)
- Personalized radiotherapy dose targeting
- Quantitative tumor burden assessment for oncology boards
- Longitudinal tracking of therapy response

Gliomas are complex, heterogeneous tumors with necrotic core, infiltrative edges, and enhancing nodules. Their segmentation demands multi-plane contextual understanding - ideally suited for AI-driven systems.

1.2 Problem Definition

Radiology departments require a system that is:

- **Automated**, with minimal manual intervention
- **Accurate**, able to localize small and irregular tumor patterns
- **Robust**, adaptable to imaging variations across scanners and centers
- **Standardized**, reducing reliance on individual expertise
- **Accessible**, deployable in standard hospital IT environments

1.3 Proposed Solution - Project Orca System

Project Orca solves these challenges via:

- **Multi-modal MRI fusion-based segmentation**
- **3D spatial processing** preserving volume integrity
- **Attention mechanisms** filtering irrelevant anatomy
- **Volumetric outputs** for quantitative decision support
- **Web-based interface** for seamless integration into workflow

The long-term clinical vision: reduce manual labeling burden while empowering radiologists with actionable data.

1.4 Objectives

Objective Category	Description
Performance	Achieve competitive Dice scores vs BraTS benchmarks
Automation	End-to-end reproducible workflow from raw input to results
Clinical Integration	Provide actionable volumetrics and overlays in a user-friendly interface
Reliability	Improve generalization and minimize false segmentation
Interpretability	Document reasoning behind decisions where possible

2. Literature Review

2.1 Computational Approaches - Historical Context

Generation	Methods	Limitations
Classical (Pre-AI)	Thresholding, atlas-mapping	Poor generalization; weak boundary localization
Machine Learning	Patch-based classification	Requires handcrafted features, limited volumetric context
Deep Learning (Current)	Fully convolutional 2D/3D architectures	Data-hungry; limited explainability

2.2 Modern Deep Learning Models for Medical Segmentation

- **U-Net:** Enabled breakthrough in biomedical segmentation efficiency

- **V-Net:** Pioneered volumetric learning
- **Attention U-Net:** Directs spatial focus to high-priority tumor regions
- **Transformer-based models:** Capture long-range semantic relationships

2.3 Commercial Clinical Solutions

Although notable industry adoption exists, high system cost and proprietary limitations restrict usage:

Platform	Capability	Adoption Barrier
Brainlab	Surgical workflow integration	Expensive system licensing
icometrix	Longitudinal tracking	Integration complexity
Quibim	Radiomics and quantitative reports	High-configuration hardware
Aidoc	Emergency triage, anomaly detection	Limited tumor-specific analysis

2.4 Key Gaps in Research & Deployment

1. **Accuracy imbalance** across sub-regions (ET weakest)
2. **Domain shift issues** when applied to real hospital data
3. **Latency constraints** in operational environments
4. **Explainability requirements** to achieve clinician trust

Project Orca is designed to fill these gaps through expanded training and smart deployment architecture.

2.5 Benchmarks and Datasets

The **Brain Tumor Segmentation (BraTS) Challenge dataset** is the universally recognized gold standard for this task. It provides thousands of high-quality, multi-modal MRI scans with manually annotated ground truth segmentations from multiple institutions. It is the primary resource for training, validating, and benchmarking models in a fair and reproducible manner. Standard evaluation metrics include the **Dice Similarity Coefficient (DSC)** and **Intersection over Union (IoU)**, which measure overlap, and the **Hausdorff Distance**, which measures boundary distance.

3. Dataset and Preprocessing

3.1 Dataset Origin - BraTS-GLI Challenge Datasets (2023 for Training, 2024 for Testing)

These scans come from the BraTS Glioma Challenge, a global competition where researchers build AI models to detect and segment brain tumors. The challenge provides high-quality, expert-labeled MRI data from many hospitals, ensuring a standardized and diverse dataset for fair comparison and real-world reliability.

For this project, two official BraTS-GLI datasets were used:

- Training Data:

BraTS-GLI (2023) -

ASNR-MICCAI-BraTS2023-GLI-Challenge-TrainingData.zip

Contains 1,251 samples

Source: <https://www.synapse.org/Synapse:syn51514132>

- Testing Data:

BraTS-GLI (2024) - BraTS2024-BraTS-GLI-AdditionalTrainingData.zip

First 150 examples from the dataset are selected for testing

Source: <https://www.synapse.org/Synapse:syn64314352>

Regulatory Compliance: - Fully de-identified / anonymized dataset - Research-only license (**CC-BY-NC 4.0**)

Professional radiologists have labeled sub-regions:

- **Enhancing core:** The bright, active part of the tumor seen after contrast injection.
- **Necrotic region:** The dark, dead tissue at the center of the tumor.
- **Edema/Swelling boundary:** The swollen, fluid-filled area around the tumor.

Modalities exploited:

Modality	Biological Signal Target
T1	Macro-anatomical structure

T1ce	Breakdown of blood-brain barrier → enhancement
T2	Water-rich structures
FLAIR	Edema visualization by CSF suppression

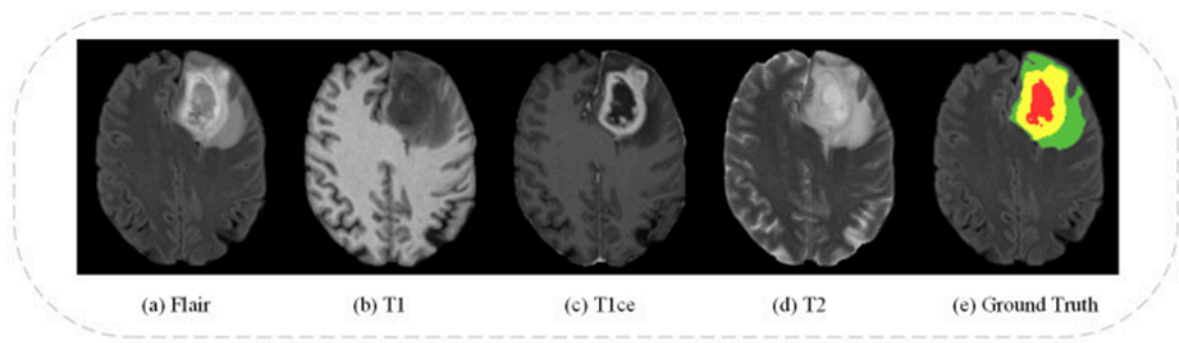


Figure: Multi-Modal MRI Inputs

3.2 Exploratory Data Analysis (EDA)

Significant observations include:

- All 3D scans were consistent in dimension (240x240x155) and voxel spacing (1.0, 1.0, 1.0)
- Variability in intensity histograms → scans showed large differences in voxel intensity ranges across patients and modalities.
- Structural regions with differing noise characteristics → visual inspection confirmed that although dimensions and voxel spacing are consistent, different modalities exhibit different contrast and noise patterns that must be normalized.
- Tumor shape diversity → tumors ranged from small, compact regions to large, irregular, and diffuse infiltrations, reinforcing the need for multi-scale feature learning.
- Presence of patients missing certain tumor sub-regions → although all scans contain tumors, not every sub-class (ET, TC components) is present in every case. The model must handle absent classes without bias.

3.3 Preprocessing Pipeline

A fully automated preprocessing pipeline was developed using the MONAI framework. This pipeline is a critical component of the project's reproducibility and robustness. For every patient, the following transformations were applied:

1. `LoadImaged`: Loads the four `.nii.gz` MRI files and the segmentation mask.
2. `ConvertToMultiChannelBasedOnBratsClassesd`: This step transforms the original single-channel segmentation mask where each voxel is labeled as 1 (necrotic core), 2 (edema), or 4 (enhancing tumor) into a three-channel binary mask, one channel for each clinically relevant region used in the BraTS challenge:
 - Whole Tumor (WT): includes all tumor-related tissues (labels 1, 2, and 4).
 - Tumor Core (TC): includes only the solid parts of the tumor (labels 1 and 4), excluding edema.
 - Enhancing Tumor (ET): includes only the enhancing region (label 4).

Each channel becomes a binary map (0/1) indicating the presence or absence of that specific region at each voxel. This representation is essential because most BraTS models are trained to predict these three composite regions rather than the raw label map.

3. `Spacingd`: Resamples all 3D volumes to a uniform isotropic voxel spacing of (1.0, 1.0, 1.0) mm. This ensures that the physical size of anatomical features is consistent across all scans.
4. `ScaleIntensityRanged`: Normalizes the voxel intensity values of the MRI scans to a standard floating-point range of [0.0, 1.0], a crucial step for stable neural network training.
5. `CropForegroundd`: Automatically crops the image to remove large areas of empty background/air, focusing computational resources on the brain tissue.
6. `Resized`: Resizes all cropped volumes to a fixed spatial dimension of (128, 128, 128). This creates uniform input tensors required by the deep learning model.

3.4 Multi-Modal Data Fusion

The project employed an early fusion strategy. After preprocessing, the four MRI modalities (T1, T1ce, T2, FLAIR) were stacked along the channel dimension, creating a single 4-channel, 3D tensor of shape (4, 128, 128, 128). This tensor served as the direct input to the model.

3.5 Data Splitting

The preprocessed dataset contained 1251 MRI samples. These were divided into an 80% training set and a 20% validation set, using a fixed random seed (RANDOM_STATE = 42) to ensure that all experiments remain fully reproducible.

- Total samples: 1251
- Training set (80%): 1000 samples
- Validation set (20%): 251 samples

For testing, we used an additional 150 completely unseen samples from the BraTS dataset. These test cases were not included in the 80/20 split, ensuring an unbiased evaluation of the model's generalization performance on new, real-world data.

4. Model Development and Experiments

4.1 Model Architecture - 3D Attention U-Net

For this project, we used a 3D Attention U-Net, implemented with the MONAI medical-imaging framework. This architecture is widely used in medical image segmentation because:

- It processes the MRI scans as full 3D volumes, not as separate 2D slices, allowing the model to understand how tumor structures continue across different layers of the brain.
- Skip connections in the U-Net help combine high-level features (e.g., “tumor-like texture”) with low-level details (e.g., edges and boundaries).
- Attention gates act like a spotlight, helping the network focus on the most important regions of the scan (the tumor), while ignoring irrelevant background information.

The model was built with five levels of downsampling and upsampling, using channel depths of (16, 32, 64, 128, 256).

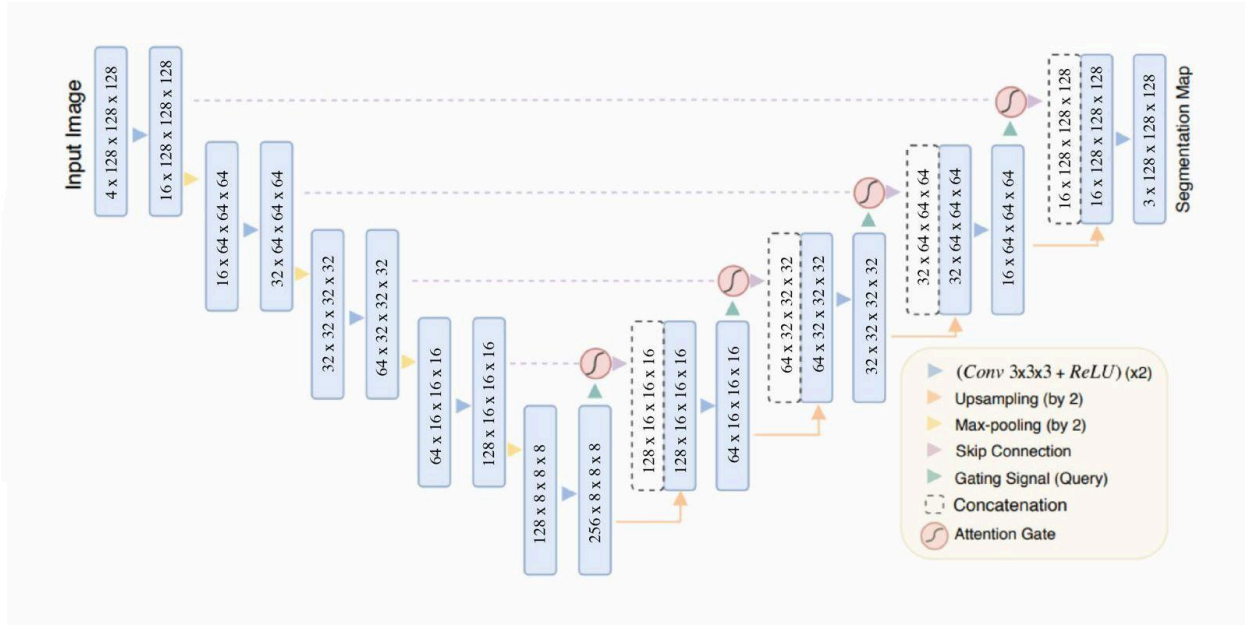


Figure: 3D Attention U-Net Architecture Diagram

This configuration provided a good balance between accuracy and computational cost.

4.2 Hyperparameter Tuning and Experimental Trials

Before finalizing the training setup, we ran several experiments to determine the best design choices.

4.2.1 Initial Experiments

We first explored:

- K-Fold Cross-Validation
 - Produced slightly better but not significantly different results
 - Required $5\times$ the compute time $\rightarrow$ not justified for our hardware
- 3D Patch-Based Training
 - Faster but misleading
 - The model became good at predicting background (which is 99% of the volume)
 - Failed to properly learn tumor structures

Because of these limitations, we proceeded with full-volume training on a standard 80/20 split.

4.2.2 Hyperparameter Sweep

To identify the most effective configuration for stable training and high-quality tumor segmentation, a systematic hyperparameter sweep was conducted. The goal of this sweep was to understand the sensitivity of the model to learning rate, network capacity, regularization strength, and the balance between Dice and BCE components in the loss function. Each experiment was run on the same 80/20 train–validation split to ensure strict comparability.

The following parameters were varied:

Parameter	Values Tested
Learning Rate (lr)	5e-05, 0.0001, 0.0005
Batch Size	1, 2
Patience (Early Stop)	7
Gradient Clipping (clip_grad)	0.5, 1.0, 2.0
Weight Decay	1e-05, 1e-04
Encoder-Decoder Channels	(8,16,32,64,128), (16,32,64,128,256), (32,64,128,256,512)
Dice–BCE Loss Weights	(0.5,0.5), (0.3,0.7), (0.7,0.3)

During the sweep, learning rate and network capacity emerged as the two most influential factors. Very small networks (starting with 8 channels) suffered from underfitting, especially for the Enhancing Tumor (ET), while extremely large networks (starting with 32 channels) increased GPU memory usage without producing a notable accuracy gain. Similarly, higher learning rates (5e-04) caused unstable training dynamics, while lower rates (5e-05) slowed convergence.

Batch size also influenced performance: a batch size of 1 led to noisy gradient updates and unstable validation curves, whereas a batch size of 2 provided a good balance between stability and memory efficiency.

After evaluating all configurations, the best-performing setup was:

- **Learning Rate:** 0.0001
- **Batch Size:** 2

- **Encoder–Decoder Channels:** (16, 32, 64, 128, 256)
- **Loss Weights:** Dice = 0.5, BCE = 0.5
- **Gradient Clipping:** 1.0
- **Weight Decay:** 1e-05

This configuration consistently produced the **smoothest training curves**, **lowest validation loss**, and **highest validation Dice scores** across regions. It was also the most computationally efficient option that did not sacrifice accuracy.

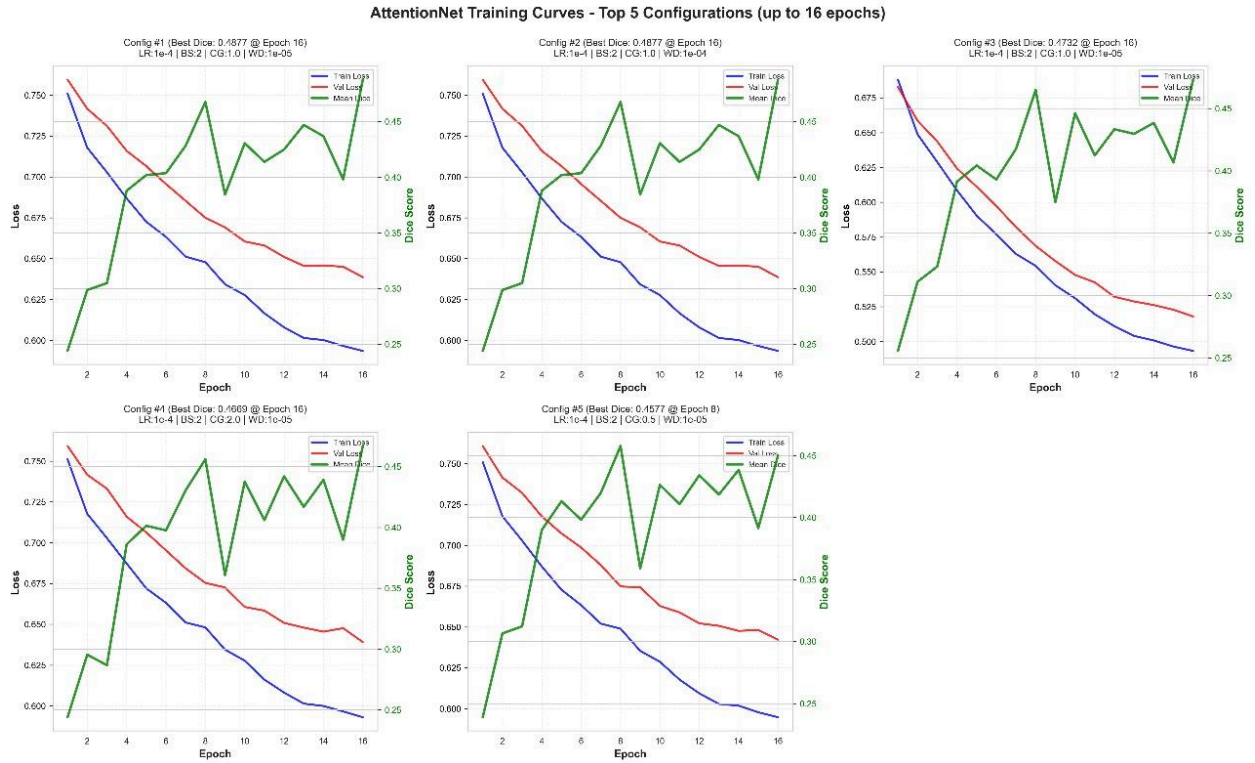


Figure: Plot: Training Curves - Top 5 Configurations

These values showed the most stable training curves and the highest validation Dice scores.

4.3 Final Training Setup

After tuning, the final configuration was fixed and used for all remaining experiments.

4.3.1 Hardware and Environment

Training was done on Kaggle using an NVIDIA Tesla P100 GPU, which supports mixed precision and allows reasonable training times for 3D models.

4.3.2 Loss Function

We used a combined Dice + BCE loss, defined as:

$$0.5 \times \text{DiceLoss} + 0.5 \times \text{BinaryCrossEntropyWithLogitsLoss}$$

- Dice Loss helps handle class imbalance (tumors are small compared to the whole scan).
- BCE stabilizes training by penalizing voxel-level mistakes.

4.3.3 Optimizer and Learning Rate Schedule

- Optimizer: AdamW
- Learning Rate: $1e-4$
- Scheduler: Polynomial Decay

AdamW helps prevent overfitting by applying better weight decay control.

4.3.4 Training Procedure

- Batch Size: 2
- Epochs: 93 (determined by early stopping)
- Early Stopping: Stop if validation Dice does not improve for 7 epochs
- Mixed Precision (AMP): Enabled to speed up training and reduce memory use

4.4 Regularization and Data Augmentation

To improve generalization and prevent overfitting, we applied:

- Random flips (left-right, up-down)
- Random rotations
- Intensity scaling

All using MONAI's Rand* transforms. These augmentations simulate real-world variations in MRI acquisition.

We also applied:

- Weight Decay (via AdamW)
- Intensity Normalization using MONAI's ScaleIntensityRanged

These ensure stable learning across diverse patients and scanners.

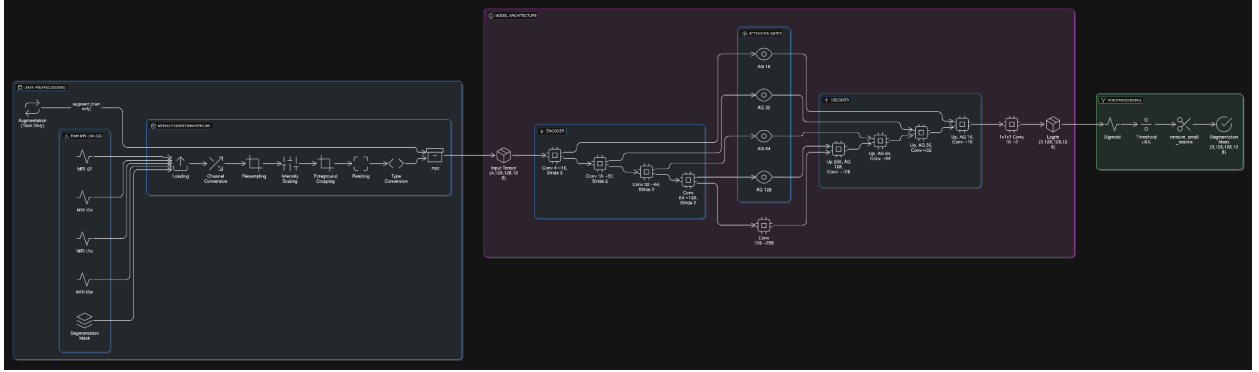


Figure: End-to-End Segmentation Pipeline: From Raw MRI Inputs to Post-Processed Tumor Masks

4.5 Results

4.5.1 Training and Validation Performance

The model showed a stable learning trend: training loss steadily decreased across epochs, while validation loss decreased initially and then fluctuated, indicating mild overfitting after prolonged training. Despite this, validation Dice scores improved consistently over the course of training.

Across the training run, the **best-performing checkpoints** marked major improvements in segmentation accuracy. Validation Dice scores for the three BraTS subregions - **Whole Tumor (WT)**, **Tumor Core (TC)**, and **Enhancing Tumor (ET)** - increased substantially compared to early epochs.

The best checkpoint (Epoch 93) achieved:

- **Dice (WT): ~0.87**
- **Dice (TC): ~0.80**
- **Dice (ET): ~0.67**

These results reflect strong segmentation performance, especially for WT and TC, which are typically easier to identify than ET due to clearer boundaries and larger regions.

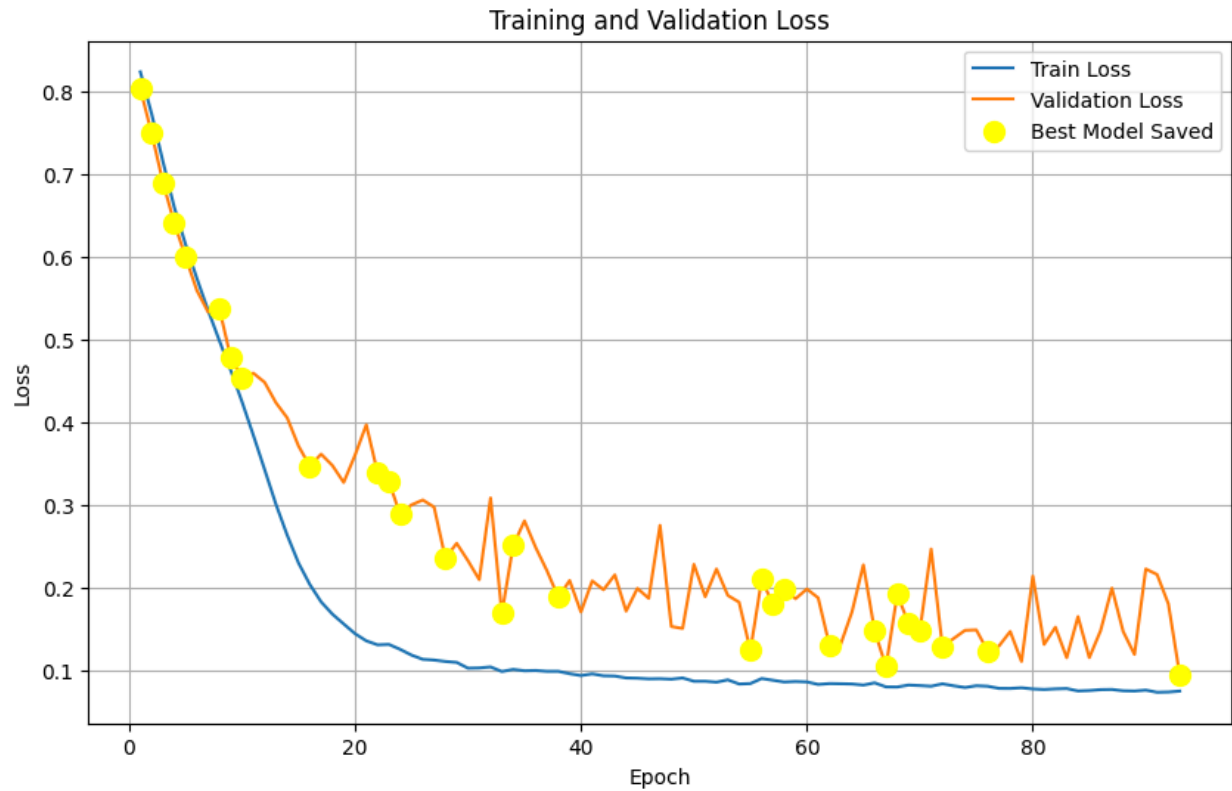


Figure: Plot: Training vs Validation Dice Score Curve

4.5.2 Validation Set Visualization

To qualitatively assess learning progression, segmentation outputs were visualized at key checkpoints (Epochs 22, 33, 55, 67, and 93).

Across epochs, the model increasingly captured:

- More complete tumor boundaries
- Clearer differentiation between WT, TC, and ET
- Fewer false positives in non-tumor regions

By the later epochs, the predicted masks closely resembled the ground truth annotations, demonstrating that the model effectively learned spatial tumor patterns.

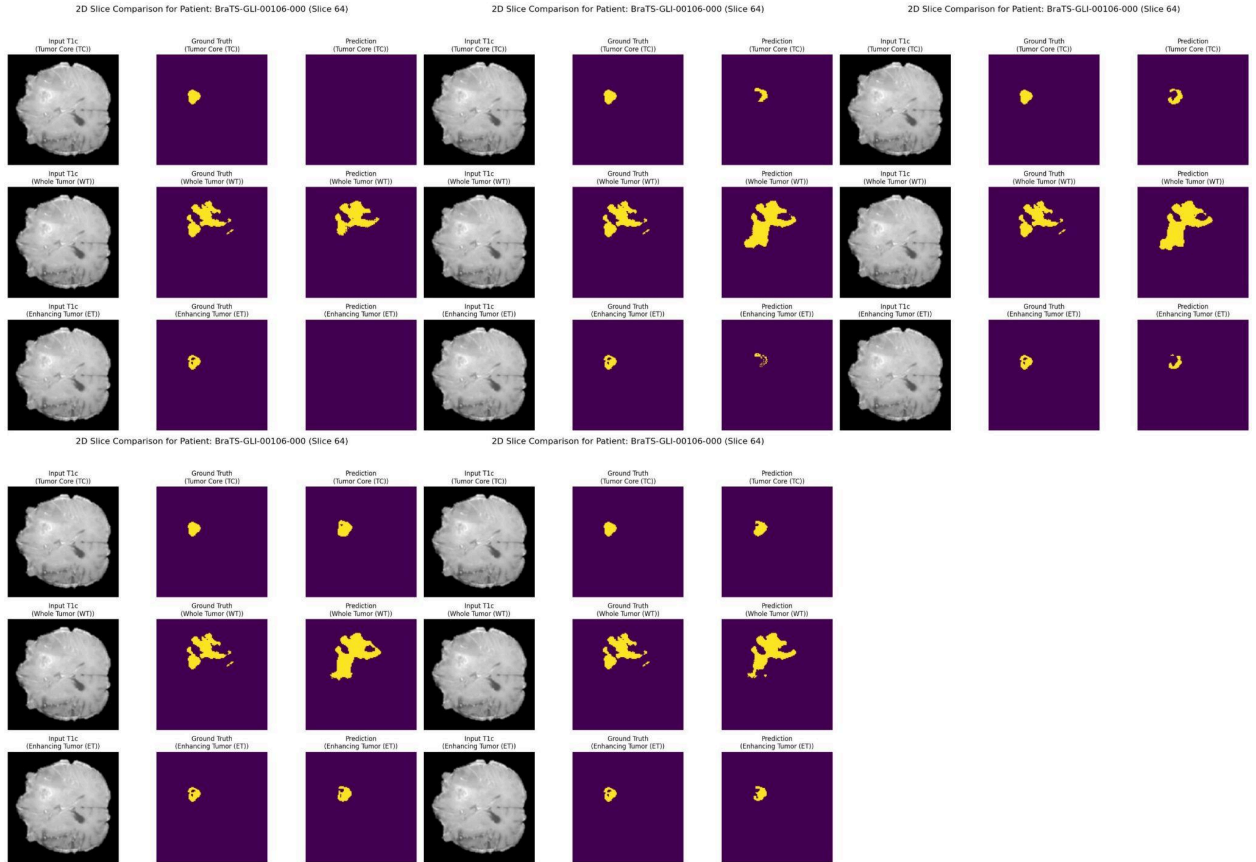


Figure 5.2: Segmentation outputs at five training milestones (Epochs 22, 33, 55, 67, and 93), arranged left to right across two rows.

4.5.3 Additional Evaluation on a Small Validation Test Set

To further test generalizability, three validation samples were evaluated.

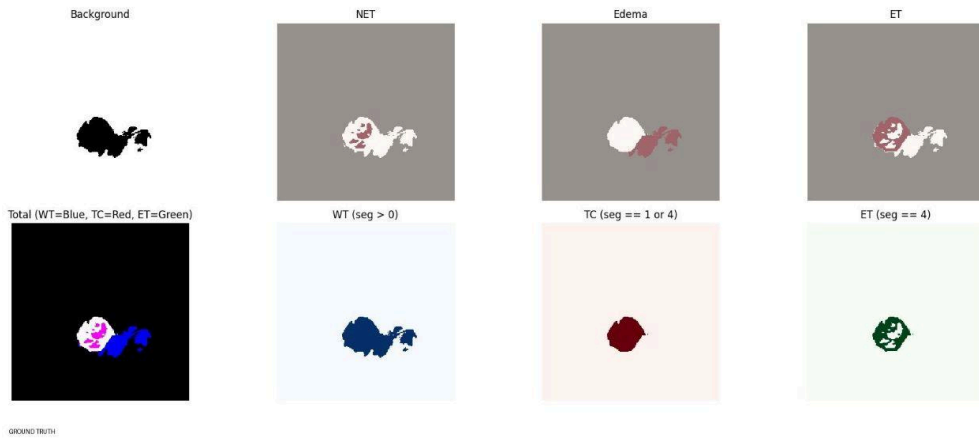
For each case, we compared:

- **Ground truth annotations** (NET, Edema, Enhancing Tumor)
- **Derived masks** (WT, TC, ET)
- **Model predictions** (raw outputs and post-processed outputs)

Post-processing helped remove small disconnected regions (“lesions”), producing cleaner and more anatomically coherent segmentation masks. In most cases, post-processed predictions aligned more closely with the ground truth, confirming the benefit of the refinement step.

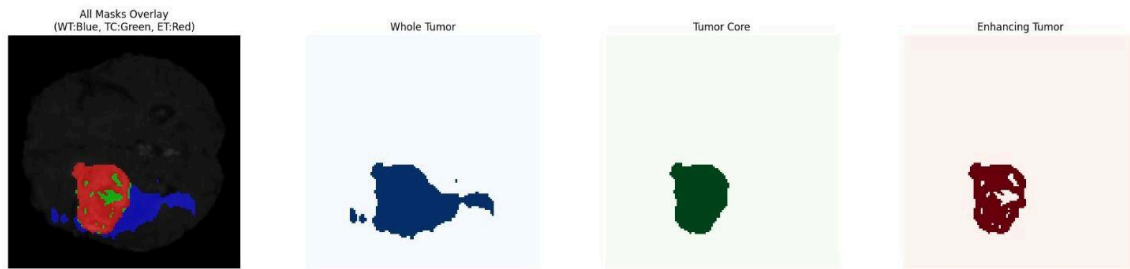
GROUND TRUTH RAW + DERIVED MASKS

BraTS-GLI-00132-000 — Raw Labels (top) and Derived Masks (bottom)
Slice 64



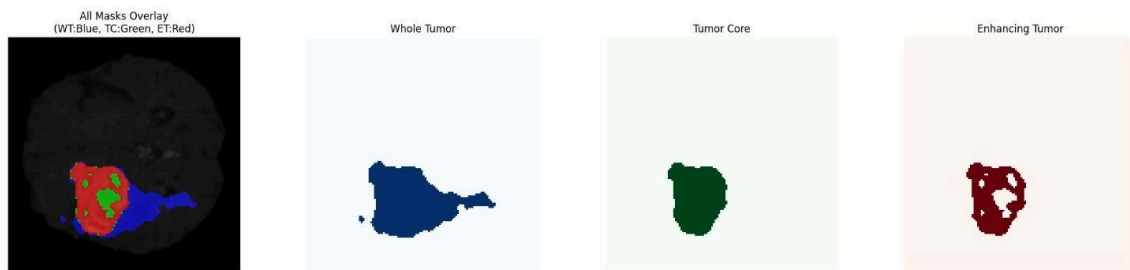
GROUND TRUTH

Model Prediction for Patient: BraTS-GLI-00132-000 (Slice 64)



PREDICTION WITH LESIONS

Model Prediction for Patient: BraTS-GLI-00132-000 (Slice 64)



PREDICTION WITHOUT LESIONS

Model Prediction for Patient: BraTS-GLI-00132-000 (Slice 64)



Figure: Validation Set Prediction (raw + post-processed)

Summary

Overall, the model demonstrated:

- Consistent improvement in Dice scores
- Strong qualitative segmentation performance
- Good generalization on unseen validation cases
- Clear benefit from simple post-processing

This performance indicates that the **3D Attention U-Net is well-suited for brain tumor segmentation**, especially when complemented by appropriate checkpointing and post-processing strategies.

5. Evaluation & Analysis

The final Attention U-Net model, trained for 93 epochs, was evaluated on an **unseen test set of 150 BraTS patient scans**. This section summarizes the quantitative performance, qualitative observations, and key error patterns identified during the analysis.

5.1 Quantitative Results

5.1.1 Dice Similarity Coefficient (DSC)

The Dice score was the primary evaluation metric. It measures how much the predicted tumor region overlaps with the ground truth mask (1 = perfect overlap, 0 = no overlap).

Dice Score Summary (150 Test Cases)

Label	Mean	Std	Median	Min	Max
Tumor Core (TC)	0.5947	0.3489	0.7515	0.0000	0.9537
Whole Tumor (WT)	0.6892	0.3140	0.8393	0.0000	0.9544
Enhancing Tumor (ET)	0.5079	0.3267	0.6373	0.0000	1.0000
Mean Dice per Patient	0.5973	0.3057	0.7183	0.0000	0.9243

WT achieved the highest performance, while **ET was the difficult**, showing more complete failures.

Clinical Interpretation:

In a clinical setting, Dice scores above **0.85** for WT and TC typically indicate segmentation reliable enough for treatment planning, surgical navigation, and volumetric progression monitoring. In our results, **high-performing WT cases (Dice ~0.9)** fall within this clinically acceptable range, showing strong potential for decision support. However, the **moderate ET performance (mean Dice ~0.51)** implies that the model may fail to reliably detect active tumor regions that radiologists use to assess malignancy progression and therapy response. ET under-segmentation can **underestimate disease aggressiveness**, potentially impacting radiotherapy targeting or clinical trial eligibility. Thus, while WT and TC predictions provide useful volumetric context, **ET predictions should not be used directly for clinical decisions without human review.**

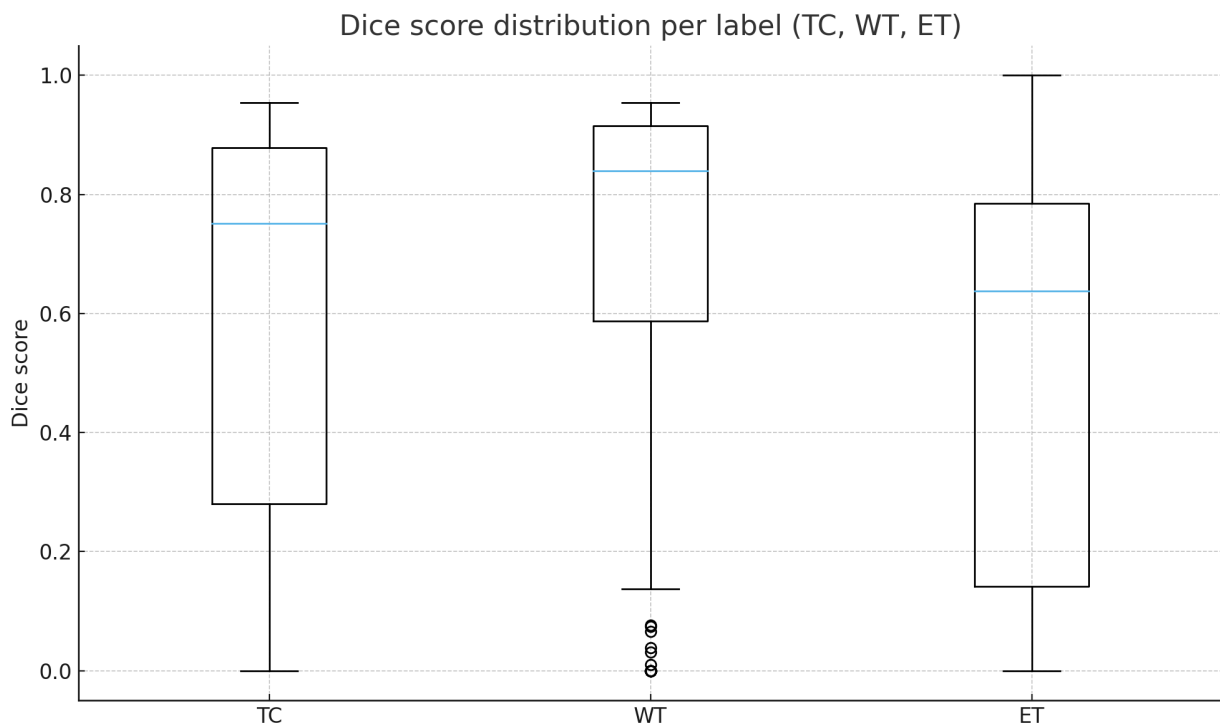


Figure: Distribution of Dice scores across tumor regions.

5.1.2 Volume Correlation (Ground Truth vs Model Prediction)

To check whether predicted tumor volumes follow real tumor size trends, Pearson correlation was computed.

Region	Pearson r
Tumor Core (TC)	0.9077
Whole Tumor (WT)	0.8786
Enhancing Tumor (ET)	0.8033
Mean Dice	0.8923

The model consistently tracks **relative tumor sizes**, though absolute values differ.

5.1.3 Linear Regression Analysis

To check calibration:

Label	Slope	Intercept	R ²
Tumor Core (TC)	0.9314	-812.34	0.8239
Whole Tumor (WT)	0.8678	-1883.73	0.7720
Enhancing Tumor (ET)	0.6677	-522.34	0.6454
Mean Dice	0.8892	-5724.07	0.7963

All slopes < 1 → systematic under-prediction of tumor volumes.

ET shows the weakest consistency (lowest R²), confirming higher difficulty.

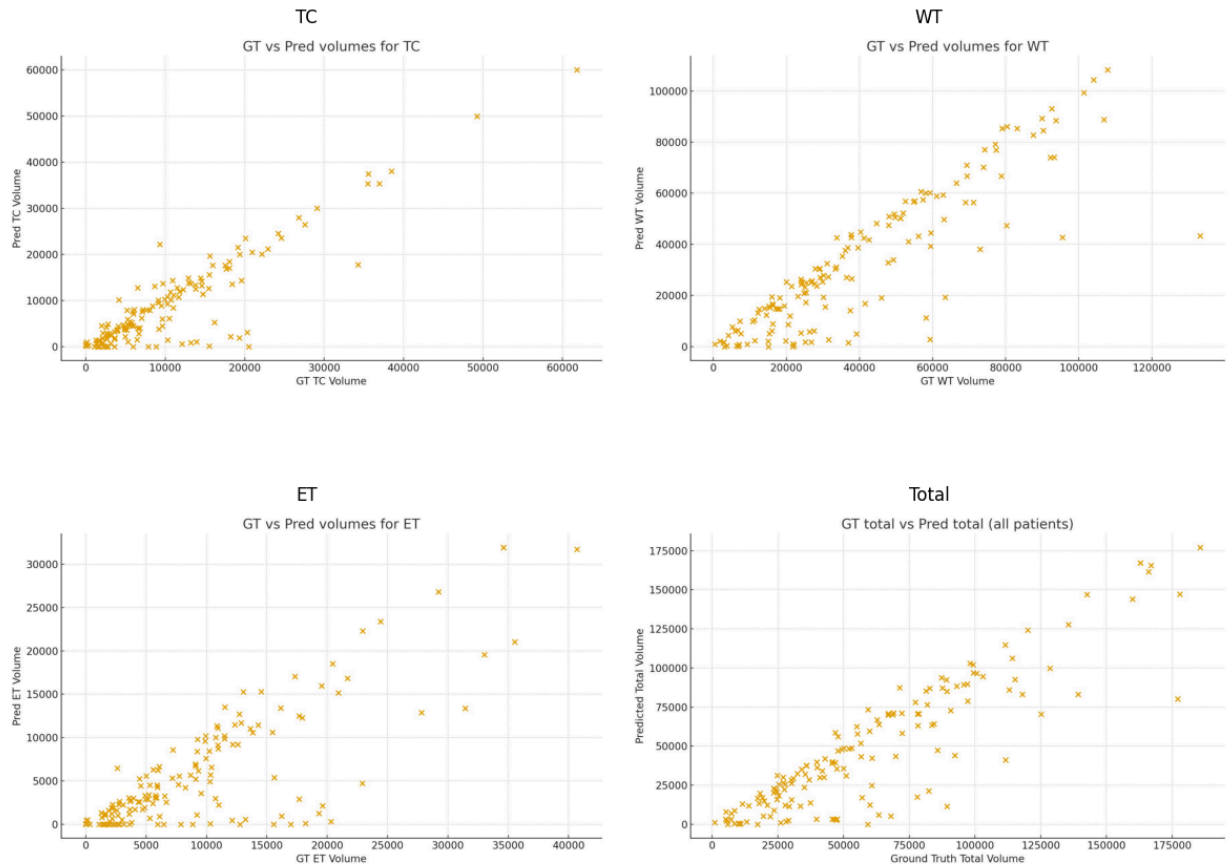


Figure: Predicted vs ground-truth tumor volumes.

5.1.4 Missed Detections (Predicted Volume = 0, GT > 0)

Label	Missed count (patients)
Tumor Core (TC)	12
Whole Tumor (WT)	3
Enhancing Tumor (ET)	22

ET was completely missed in **22 out of 150** cases - a strong indicator of class imbalance and difficulty detecting small enhancing regions.

5.1.5 Performance Comparison with BraTS 2023 Leaderboard

To contextualize our model's performance against current benchmarks, we compared the validation-set Dice scores with the BraTS 2023 leaderboard averages and the top-performing method (nnUNet).

Metric	Our Model (Validation)	BraTS 2023 Average	Top Method (nnUNet)	Gap
Whole Tumor (WT) Dice	0.8725	0.860	0.910	+1.5%
Tumor Core (TC) Dice	0.8033	0.810	0.867	-0.8%
Enhancing Tumor (ET) Dice	0.6679	0.780	0.850	-14.4%
Overall Mean Dice	0.7812	0.817	0.876	-4.4%

Interpretation:

Our model performs competitively on WT (slightly above the BraTS mean) and reasonably well on TC, but falls substantially behind on ET consistent with the earlier observed difficulty detecting small enhancing regions. The gap in ET performance is the dominant driver of the overall mean Dice difference.

5.2 Qualitative Analysis

Visual inspection of segmentation overlays provided additional insights:

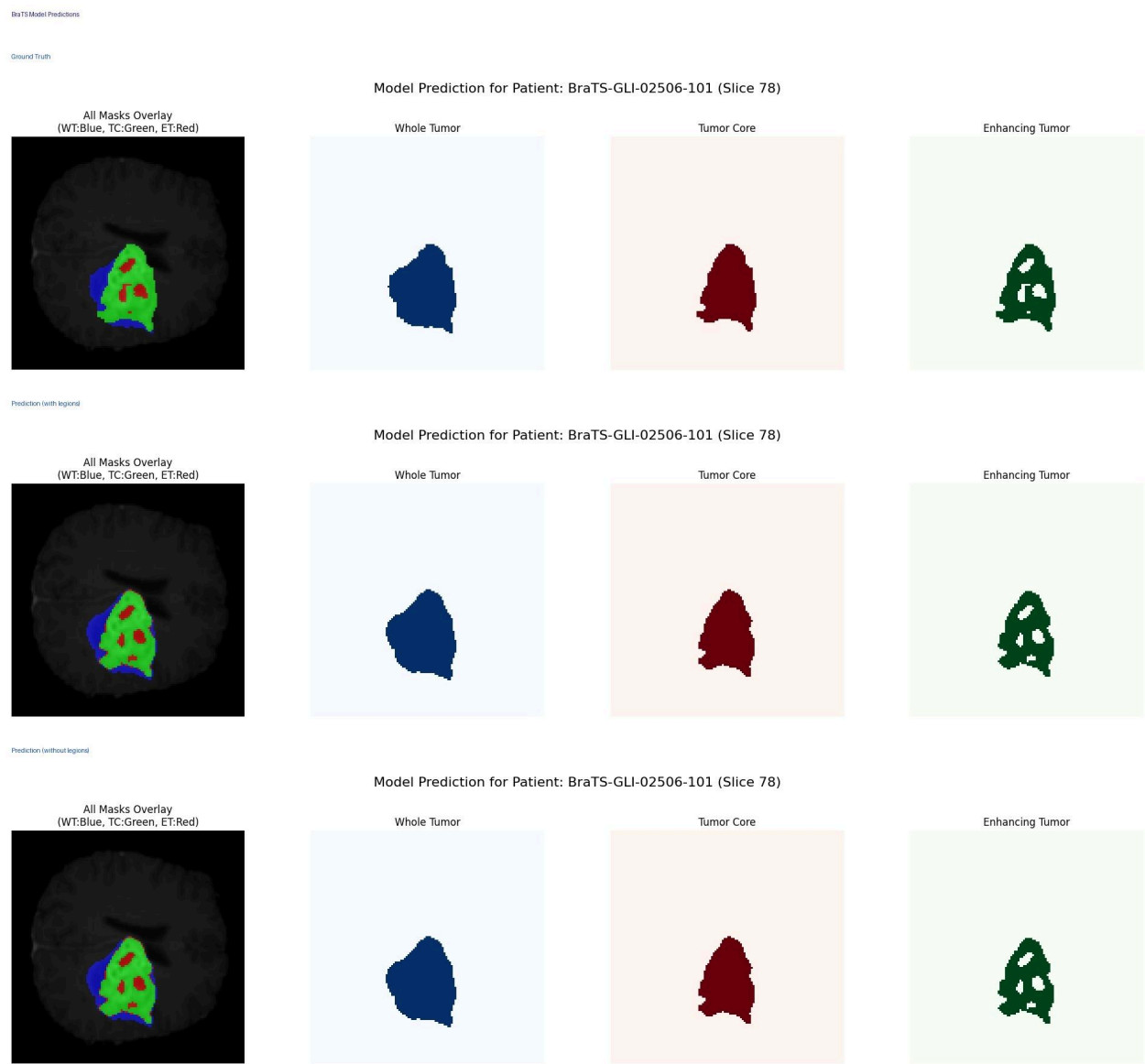
5.2.1 Successful cases

Large, well-defined tumors produced high Dice scores (>0.9). Predicted masks were almost identical to ground truth.

Test Sample "BraTS-GLI-02506-101"

Volumetric and Accuracy Analysis

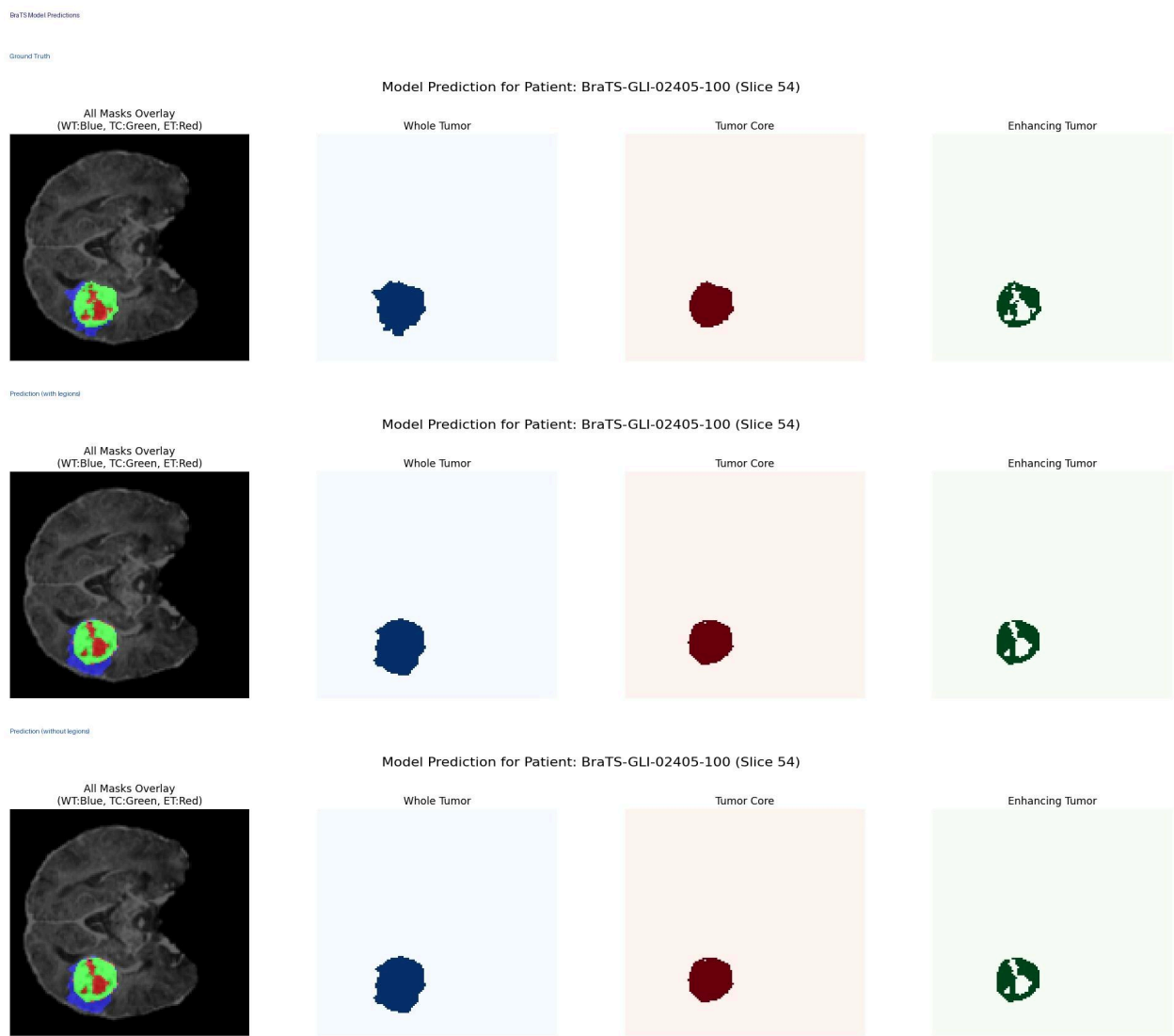
Tumor Component	Ground Truth Volume	Predicted Volume	Dice Score (Accuracy)
Tumor Core (TC)	26805.00	27970.00	0.9317
Whole Tumor (WT)	49522.00	51834.00	0.9465
Enhancing Tumor (ET)	22927.00	22288.00	0.8947



Test Sample "BraTS-GLI-02405-100"

Volumetric and Accuracy Analysis

Tumor Component	Ground Truth Volume	Predicted Volume	Dice Score (Accuracy)
Tumor Core (TC)	8419.00	8817.00	0.9117
Whole Tumor (WT)	12529.00	14535.00	0.8841
Enhancing Tumor (ET)	6093.00	6636.00	0.8060



5.2.2 Challenging cases

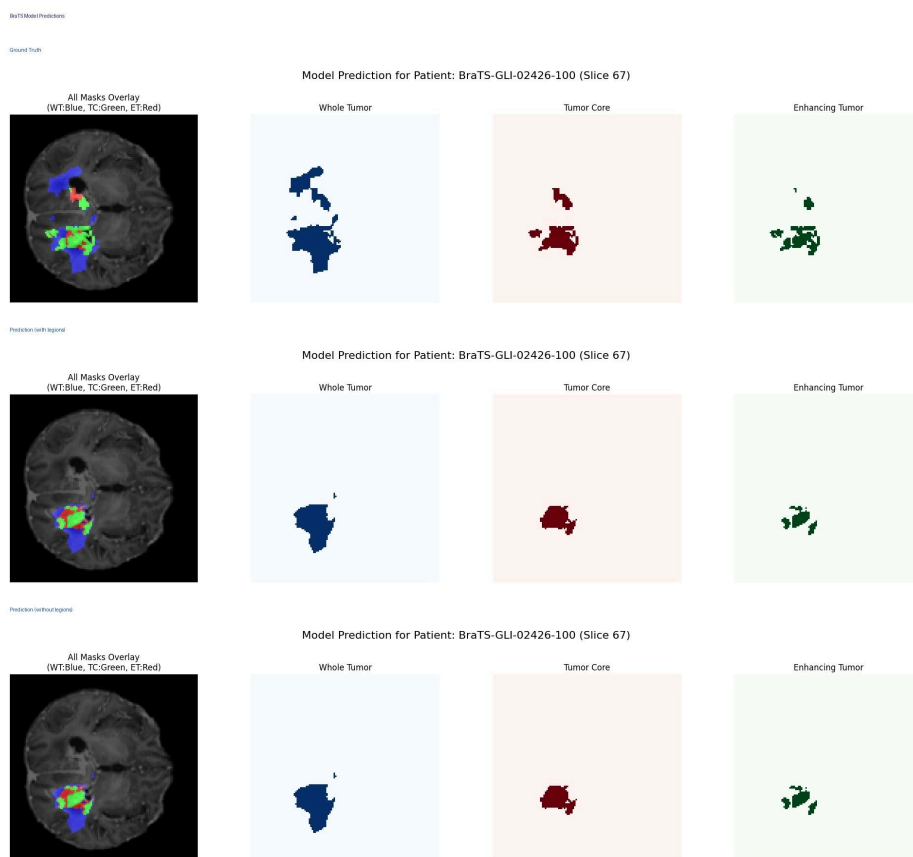
Small or low-contrast enhancing regions frequently resulted in:

- fragmented predictions
- noisy false positives
- complete misses (ET = 0)

Test Sample "BraTS-GLI-02426-100"

Volumetric and Accuracy Analysis

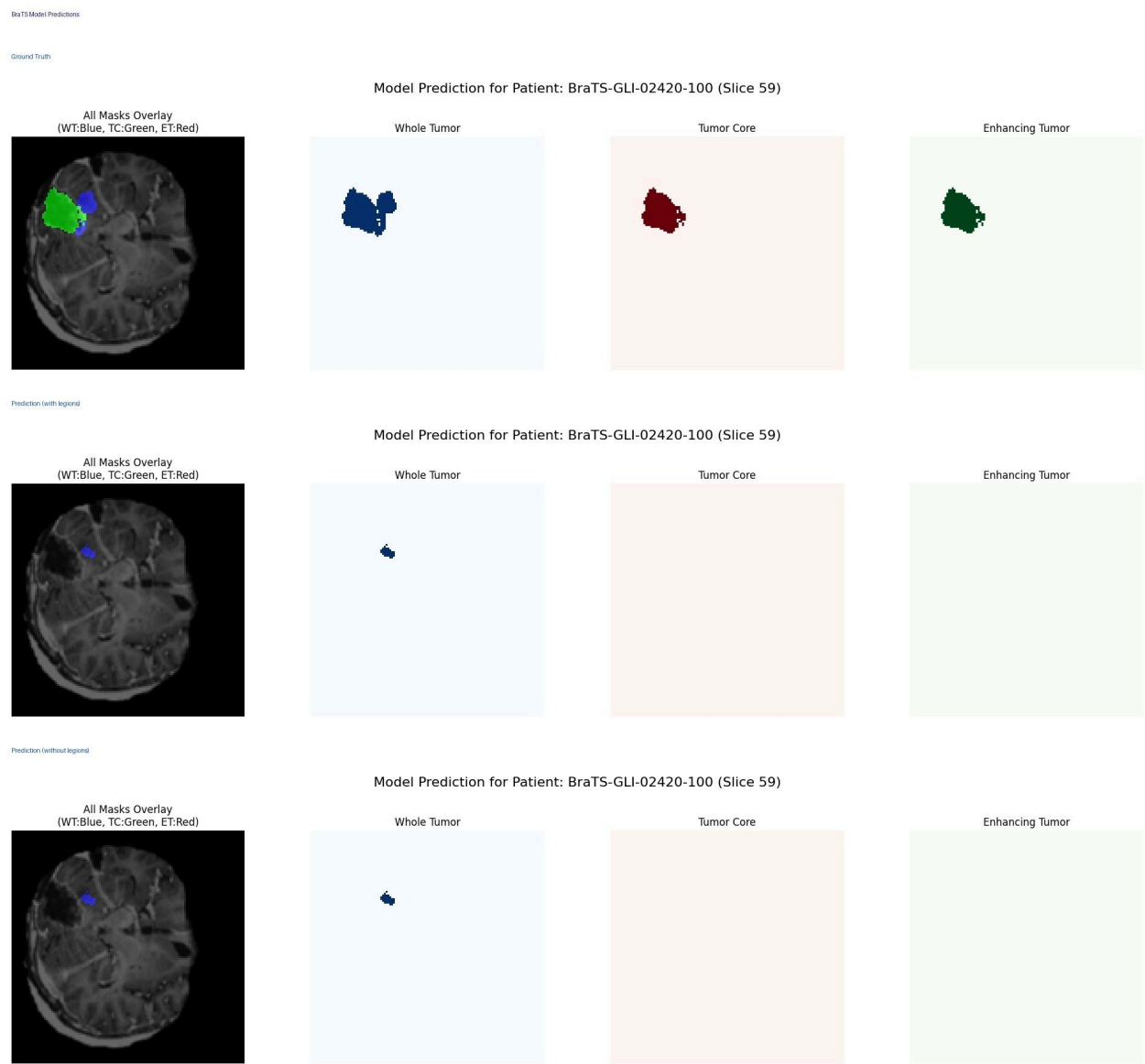
Tumor Component	Ground Truth Volume	Predicted Volume	Dice Score (Accuracy)
Tumor Core (TC)	7011.00	6168.00	0.5899
Whole Tumor (WT)	37846.00	26524.00	0.7535
Enhancing Tumor (ET)	5151.00	2907.00	0.5905



Test Sample "BraTS-GLI-02420-100"

Volumetric and Accuracy Analysis

Tumor Component	Ground Truth Volume	Predicted Volume	Dice Score (Accuracy)
Tumor Core (TC)	5968.00	0.00	0.00
Whole Tumor (WT)	7583.00	5073.00	0.07
Enhancing Tumor (ET)	5968.00	0.00	0.00



5.2.3 Failure Case Categorization

Beyond individual examples, failure cases were grouped into three recurrent categories:

1. Complete Misses (ET = 0 or TC = 0)

Observed frequently in ET (22 cases), and occasionally in TC. These arise when enhancing regions are very small (<2 ml) or poorly contrasted.

Clinically, such misses can lead to **underestimation of active disease**, making the case unsuitable for automated treatment planning without radiologist correction.

2. Fragmented or Noisy Predictions

Common in diffuse or infiltrative tumor types where boundaries are ambiguous. Fragmentation manifests as multiple scattered components instead of a contiguous region.

Clinically, this reduces trust in volumetric measurements, especially in longitudinal follow-up where consistency is essential.

3. Under-Segmentation of Large Tumors

Cases where the model captures the central bulk but fails to include irregular peripheral extensions. This directly contributes to the observed systematic volume under-prediction.

From a clinical perspective, this can impact **dose-planning margins** in radiotherapy, where accurate outer boundaries are critical.

5.3 Error Analysis

Key Patterns

1. **Systematic Under-prediction:** Regression slopes < 1 and negative intercepts show that the model consistently outputs smaller volumes than ground truth.
2. **Class Imbalance Effects:** ET is small and sparse → leads to higher false negatives and high variance.
3. **Architectural Constraints:** Attention U-Net struggles with very small or irregular subregions likely due to limited receptive field at fine levels.
4. **Post-processing Trade-off:** Threshold-based removal reduces noise but may eliminate true small ET regions.

Clinical Risk Summary:

Each error pattern carries distinct clinical implications.

- **Systematic under-segmentation** may lead to underestimation of tumor burden when tracking progression.
- **Missed ET regions** pose the highest risk, as enhancing regions are a key marker of active disease and therapy response.
- **Fragmented predictions** reduce usability in radiotherapy planning where precise contours are required.
Thus, while the model provides valuable volumetric cues, its outputs - especially ET and boundary delineation - require **radiologist validation** before incorporation into clinical workflows.

5.4 Runtime & Memory Analysis

Although detailed benchmarking was not performed, the practical runtime characteristics of the final model were assessed qualitatively during testing and deployment.

Model Size and Memory Requirements

The trained Attention U-Net model occupies approximately **72 MB**, making it lightweight compared to many modern 3D medical segmentation architectures. Because of this small footprint:

- **RAM requirements are minimal**, and the model can comfortably be loaded on typical consumer-grade machines.
- No specialized GPU hardware is required for inference.
- The training is done on Kaggle P100 GPU which have 16gb of GPU Ram and 29GB of CPU Ram.

Inference (CPU-Based Execution)

- **Inference can be performed entirely on a normal CPU**, without the need for a GPU.
- The runtime per scan generally ranges from **a few seconds to a couple of minutes**, depending on:
 - CPU speed, I/O disk speed (MRI volumes are large), and device.

This behavior is expected given the small model size and the relatively compact architecture of the 3D Attention U-Net.

MRI Scan Sizes

The MRI scans themselves ($240 \times 240 \times 155$ voxels $\times$ 4 modalities) contribute more to memory usage than the model. Storage requirements vary by compression format, but typically:

- **Each patient case** is on the order of **100MB - 300MB generally**,
- Easily handled on standard systems.

Conclusion

Overall, the model is **lightweight, efficient, and easy to run**, even on everyday computers. It does not require high-end hardware, and inference remains fast enough for practical experimentation or integration into a clinical-research workflow.

6. Deployment & Documentation

6.1 System Architecture and Deployment

The final, trained model was integrated into a fully functional, end-to-end web application.

- Backend: A Flask server written in Python handles all core logic, including:
 - File uploads.
 - Execution of the MONAI preprocessing pipeline.
 - Inference with the loaded PyTorch model.
 - Post-processing of the prediction mask.
 - Generation of 2D slice images and 3D mesh data.
- Frontend: A single-page application built with HTML, CSS, and JavaScript provides the user interface. It communicates with the Flask backend via asynchronous API calls.
- Deployment: The application is containerized using Docker and is deployed on Hugging Face Spaces, making it publicly accessible for demonstration purposes. It can also be run locally with minimal setup.

6.2 User Workflow

The application provides an easy workflow for tumor segmentation:

1. Load Data: The user can either upload their own set of four NIfTI files (FLAIR, T1, T1ce, T2) or select a pre-loaded example case.
2. Run Prediction: With a single click, the user can trigger the entire inference pipeline on the backend.
3. Visualize Results: The system presents the results in two interactive viewers:
 - A 2D Slice Viewer to inspect the input modalities and predicted masks on a slice-by-slice basis.
 - A 3D Viewer, powered by Three.js, which renders the brain and the predicted tumor sub-regions as interactive 3D meshes.

6.3 Documentation

Comprehensive documentation was created to ensure the project is understandable, reproducible, and maintainable. This includes:

- A Technical Document detailing the environment setup, data pipeline, model architecture, and inference process.
 - A User Guide with step-by-step instructions for using the web application.
 - Well-commented Code and README files for both the training pipeline and the inference application.
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7. Conclusion and Future Work

7.1 Conclusion

Project Orca successfully achieved its primary objective of creating an end-to-end deep learning system for automated brain tumor segmentation. The 3D Attention U-Net model demonstrates promising performance, particularly in delineating the whole tumor and tumor core. The project's main contribution is the creation of a complete, deployable tool that bridges the gap from a raw MRI scan to an interactive, quantitative analysis, showcasing the potential of deep learning to enhance clinical workflows. While the model shows strong capabilities, the evaluation also highlighted the persistent challenges in medical imaging, particularly the accurate segmentation of small and variable sub-regions like the enhancing tumor.

7.2 Future Work

Based on the project's findings, several avenues for future improvement are proposed:

- **Advanced Model Architectures:** Explore an ensemble model, combining the predictions of the current Attention U-Net with a complementary architecture like a ResU-Net or a Vision Transformer, which could improve robustness and accuracy.
- **Class-wise Training for Raw Labels:** Future work can explore training the model on the **individual BraTS labels (necrosis, edema, enhancing tumor)** instead of directly predicting WT, TC, and ET. This could produce cleaner raw segmentations, from which WT/TC/ET can be constructed more accurately.
- **Targeted Loss Functions:** Implement more advanced loss functions, such as the Focal Tversky Loss or boundary-aware losses, to specifically address the class imbalance even better and improve the segmentation of the Enhancing Tumor.
- **Improved Post-processing:** Develop more sophisticated post-processing techniques, potentially incorporating anatomical priors to remove false positives and ensure that predictions are anatomically plausible.
- **Generalization to Real-World Data:** To bridge the gap to clinical applicability, the model should be fine-tuned and validated on datasets from different institutions and MRI scanners to improve its generalization capabilities.
- **Integration of GenAI for Reporting:** Fulfill the original secondary objective by integrating a Generative AI model (e.g., a fine-tuned LLM) to translate the quantitative volumetric outputs into a preliminary, human-readable clinical report summary.

8. References and Appendix

8.1 References

1. Baid, U., et al. (2021). *The RSNA-ASNR-MICCAI BraTS 2021 Benchmark on Brain Tumor Segmentation and Radiogenomic Classification*. arXiv:2107.02314.
2. Menze, B. H., et al. (2015). *The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS)*. IEEE Transactions on Medical Imaging, 34(10), 1993-2024.

3. Ronneberger, O., Fischer, P., & Brox, T. (2015). *U-Net: Convolutional Networks for Biomedical Image Segmentation*. Medical Image Computing and Computer-Assisted Intervention (MICCAI).
4. Oktay, O., et al. (2018). *Attention U-Net: Learning Where to Look for the Pancreas*. arXiv:1804.03999.

8.2 Project Team

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